

Notes: Goldsmith-Pinkham and Shue (JF, 2023)

The Gender Gap in Housing Returns

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1 Big picture

- **Question.** Do single women and single men earn different returns on housing investment, and if so, what explains the gap?
- **Setting.** The paper studies U.S. housing transactions from 1991 to 2017 using CoreLogic deeds data, linked property listings data, and supplementary American Housing Survey (AHS) data.
- **Core finding.** Single women earn about 1.5 percentage points lower unlevered annualized housing returns than single men. The gap becomes much larger when returns are levered through mortgage borrowing.
- **Two broad sources.** Roughly 45% of the raw return gap is explained by market timing and location: women are more likely to buy in locations and times with high prices and sell when prices are lower. The remaining gap is linked to execution prices at purchase and sale.
- **Execution-price channel.** Conditional on property fixed effects and local time trends, women buy the same property for roughly 1% to 2% more than men and sell for roughly 2% to 3% less than men.
- **Negotiation evidence.** Women list homes for less and negotiate worse discounts relative to the list price. The highest transaction prices occur when the seller is male and the buyer is female; the lowest occur when the seller is female and the buyer is male.
- **Market tightness evidence.** The gender gap shrinks toward zero in tighter housing markets, where bilateral negotiation is more likely to be replaced by quasi-auctions.
- **Economic magnitude.** The authors estimate that housing-related return differences explain about 30% of the overall gender gap in wealth accumulation at retirement.

One-sentence summary. This note summarizes the paper, which uses national deeds records, repeat-sales transactions, linked MLS listings, name-based gender classification, property fixed effects, market timing controls, AHS validation, and market-tightness tests to show that single women earn lower housing returns than single men, with the gap driven partly by where and when they transact and partly by worse purchase and sale execution prices that are consistent with a negotiation channel.

2 Incremental contribution of the paper

- **From stock-market gender gaps to housing-market gender gaps.** A large finance literature studies gender differences in stock market participation, portfolio choice, and trading. This paper shifts attention to housing, the dominant asset in the savings portfolio of many households.
- **Returns, not only prices.** Prior work documents gender gaps in purchase prices in particular settings. This paper measures realized annualized returns by linking purchase and sale transactions for the same homeowner and property.
- **National scale and long horizon.** The paper uses over 50 million U.S. housing transactions and approximately 8.9 million realized-return observations, allowing state, time, return-distribution, and demographic heterogeneity analyses.

- **Decomposition into market timing versus execution.** The paper separates the raw return gap into a market-timing component and a residual component related to the prices at which investors execute purchases and sales.
- **Direct evidence on negotiation margins.** By linking deeds to MLS listings, the paper observes list prices, sale prices, negotiated discounts, days on market, and listing agents. This allows the authors to test whether the gender gap operates through first offers and negotiated discounts.
- **Property fixed effects.** Repeat-sales data allow the paper to hold constant time-invariant property quality when comparing transaction prices, list prices, and sale prices across gender groups.
- **Alternative-mechanism tests.** The paper examines leverage, property characteristics, maintenance, upgrades, demographics, real estate agents, holding length, liquidity, and market tightness.
- **Wealth-accumulation implication.** The paper links housing-return gaps to household wealth inequality, showing that small differences in execution prices can become economically large because housing is large, leveraged, and repeatedly traded.

3 Data and measurement

3.1 CoreLogic deeds data

- The main data source is CoreLogic county deeds records covering arms-length U.S. housing transactions from 1991 to 2017.
- Each observation contains transaction date, sale price, property address, and names of buyers and sellers.
- The authors exclude nontransaction deed events such as refinancings.
- The deeds data are used to identify repeat sales of the same property and realized returns from purchase to sale.

Interpretation. The deeds data provide the scale and transaction-level detail needed to estimate realized housing returns. The key limitation is that gender and family structure are not directly reported and must be inferred from names.

3.2 CoreLogic listings data

- The paper links deeds data to CoreLogic property listings from MLS systems.
- Listings include list price, sale price, listing date, close date, days on market, property characteristics, listing agent identifiers, and textual descriptions of upgrades or renovations.
- The linked deeds-listings data are crucial for distinguishing transaction prices from list prices and negotiated discounts.

Interpretation. The listing data convert the return gap from a black-box outcome into a set of observable bargaining margins: initial list price, discount relative to list price, time to sale, and agent choice.

3.3 Gender and family-structure classification

- The authors parse buyer and seller name fields to classify each side of a transaction as single male, single female, couple, or other.
- First names are matched to gender probabilities based on external name data. Names with gender probability above 95% are classified as male or female.
- The couple category requires two individuals with both genders identified.
- The residual other category includes unidentified names, institutions, and ambiguous family structures.
- The authors screen county-years where recording conventions appear to list only one name for what are likely couple transactions.

Interpretation. The classification strategy is conservative: it sacrifices coverage to reduce misclassification. Measurement error likely attenuates estimated gender differences, and the authors validate the main result using AHS data where gender and marital status are self-reported.

3.4 Unlevered and levered returns

For property i purchased in year b and sold in year s , the unlevered annualized return is

$$r_{is} = \left(\frac{P_{is}}{P_{ib}} \right)^{1/(s-b)} - 1,$$

where P_{ib} is the purchase price and P_{is} is the sale price.

- The return sample requires that the seller at sale can be matched to the buyer at purchase.
- The authors require the same gender/family-structure group at purchase and sale.
- This restriction excludes cases such as buying as a single person and selling as a couple, or buying as a couple and selling as a single person.
- The final returns sample contains approximately 8.9 million observations.

The paper also calculates hypothetical levered returns assuming a 30-year fixed-rate mortgage with 80% loan-to-value (LTV), market interest rates at purchase, and no refinancing. This approximation is used because actual mortgage variables are often missing.

Interpretation. Unlevered returns isolate the price appreciation component. Levered returns show that small differences in purchase and sale prices become much larger for household wealth when homes are financed with mortgage debt.

4 Empirical specifications

4.1 Return regression

The baseline return regression is

$$r_{is} = \beta_1 \text{SingleFemale}_{is} + \beta_2 \text{Couple}_{is} + X'_{is}\tau + \varepsilon_{is},$$

where single male sellers are the omitted group.

- Without controls, β_1 measures the raw return gap between single women and single men.
- The control vector X_{is} is gradually expanded to include market-timing controls: sale zip-by-year-month fixed effects, purchase zip-by-year-month fixed effects, and sale-year-month by buy-year-month fixed effects.
- The comparison therefore moves from raw returns to comparisons among households buying and selling in the same local markets at the same times.

Interpretation. This specification decomposes the return gap into a market-timing component and a residual within-location-time return gap.

4.2 Transaction-price, list-price, and discount regressions

For transaction-level outcomes, the paper estimates regressions of the form

$$Y_{it} = \beta_1 \text{SingleFemale}_{it} + \beta_2 \text{Couple}_{it} + \beta_3 \text{Other}_{it} + X'_{it}\tau + \varepsilon_{it},$$

where Y_{it} may be log sale price, log purchase price, log list price, negotiated discount, probability of sale, or days on market.

- These regressions typically use property fixed effects to compare transaction outcomes for the same property across different owners or counterparties.
- They also include zip-by-year-month fixed effects to absorb local housing-market trends.
- Discounts are defined as

$$\text{Discount} = 100 \times \frac{\text{ListPrice} - \text{TransactionPrice}}{\text{ListPrice}}.$$

- A larger purchase discount benefits buyers; a larger sale discount hurts sellers.

Interpretation. These regressions identify whether the residual return gap comes from worse execution at purchase, worse execution at sale, or both.

4.3 Mechanism tests

The paper evaluates several potential mechanisms:

- **Market timing:** Are women buying in high-price locations/times and selling in low-price locations/times?

- **Negotiation:** Do women list at different prices and negotiate different discounts?
- **Liquidity:** Do female sellers sell faster or with higher probability, offsetting lower sale prices?
- **Holding length:** Does the annualized return gap decay with holding length, as expected if the gap reflects fixed execution-price differences at purchase and sale?
- **Risk and maintenance:** Do men buy riskier homes or invest more in maintenance/upgrades?
- **Demographics:** Does the gap survive controls for age, income, education, children, and ethnicity in AHS data?
- **Agents:** Does the gap survive listing-agent fixed effects?
- **Market tightness:** Does the gap shrink in tight markets where negotiation is replaced by quasi-auction bidding?

5 Identification and empirical design

5.1 What the design identifies

- The paper identifies robust conditional gender differences in realized housing returns and transaction outcomes.
- Repeat-sales property fixed effects remove time-invariant property quality from price comparisons.
- Zip-by-year-month fixed effects remove local price trends and market conditions at the transaction date.
- The return regressions with purchase and sale timing controls measure whether the gender return gap remains among households transacting in the same local markets at the same times.

Interpretation. The empirical design is strongest at showing that the return gap is not simply due to women buying different types of homes or transacting in systematically different local market conditions.

5.2 What remains observational

- The paper does not randomly assign gender, negotiation behavior, home choices, or market timing.
- It cannot fully separate negotiation ability, preferences over confrontation, advice from real estate agents, constraints created by family responsibilities, and discrimination by counterparties.
- It measures financial returns, not welfare. Women may derive different consumption value from homeownership, which is not observed.

Interpretation. The most appropriate reading is that gender is strongly associated with lower financial housing returns, and that the evidence is most consistent with market timing plus negotiation/execution-price channels.

5.3 Why holding length and market tightness are important

- If the return gap comes from a fixed purchase-price disadvantage and a fixed sale-price disadvantage, then the annualized return gap should be larger for short holding periods and shrink toward zero as holding length grows.
- If the gap comes from men buying riskier homes or maintaining homes better each year, the annualized return gap should persist even at long holding lengths.
- The paper finds that the annualized return gap decays toward zero with holding length, supporting the execution-price explanation.
- In tight markets, multiple buyers can bid simultaneously, reducing the role of bilateral negotiation. The gender gap shrinks in these markets, further supporting a negotiation channel.

6 Tables and figures

6.1 Table I and Figure 1: sample structure and baseline return gap

Read:

- Table I summarizes the full transactions sample, the listings sample, and the returns sample by gender group.
- The full sample contains over 52 million arms-length transactions. The returns sample contains approximately 8.9 million observations.
- In the returns sample, single male sellers have mean unlevered annualized returns of 8.40%, single female sellers 6.85%, and couples 6.46%.
- Figure 1 shows that the raw gender gap narrows but remains economically meaningful after adding location and timing fixed effects.
- The levered return gap is much larger than the unlevered return gap because mortgage leverage amplifies price-return differences.

Economic message: The return gap is large in raw data and remains substantial after controlling for market timing. Leverage converts modest price differences into much larger differences in household equity returns.

Table I
Summary Statistics

This table reports summary statistics for the samples used in the analysis, split by four gender groups (single male, single female, couple, and other) and also pooled. Panels A and B are split by buyer gender group, and Panels C, D, and E are split by seller gender group. Each cell is the overall mean within the relevant sample group. Panels A and C represent the full sample of all arms-length sales transactions reported in the data. Panels B and D represent the sample of sales transactions successfully matched to listings data. Panel E represents the sample of sales transactions for which we successfully match the identified gender and family structure in the sale transaction with that in the previous purchase transaction, and also match the names of the sellers with the names of the buyers in the previous transaction. See Section 1B for more details on how we identify gender groups (single female, single male, couple, and other). See Section 1A for more details on how we match the data.

	Buyer Gender Group				
	Single Male	Single Female	Couple	Other	Overall
Panel A: Full Sample					
Log(Sale Price)	11.9482	11.8821	12.1105	12.1842	12.0704
Sample Size	11,561,755	7,614,231	13,424,894	20,282,986	52,883,866
Panel B: Listing Sample					
Log(Sale Price)	12.0567	11.9758	12.2847	12.0711	12.1102
Log(List Price)	12.0739	11.9918	12.3067	12.0958	12.1308
Purchase Discount (p.p.)	2.7826	2.5422	2.7200	3.1715	2.8415
Log(Days on Market)	3.7355	3.7024	3.7856	3.7377	3.7444
Sample Size	4,751,834	3,240,324	5,236,683	5,711,909	18,940,750
	Seller Gender Group				
	Single Male	Single Female	Couple	Other	Overall
Panel C: Full Sample					
Log(Sale Price)	12.0264	11.9572	12.1414	12.0748	12.0704
Sample Size	6,092,999	5,110,755	9,960,567	31,719,545	52,883,866
Panel D: Listing Sample					
Log(Sale Price)	12.1459	12.0678	12.2737	12.0332	12.1102
Log(List Price)	12.1651	12.0914	12.2919	12.0546	12.1308
Sale Discount (p.p.)	2.7882	2.9663	2.5274	2.9744	2.8415
Log(Days on Market)	3.7176	3.6967	3.7018	3.7840	3.7444
Sample Size	2,484,028	2,423,657	4,554,616	9,478,449	18,940,750
Panel E: Returns Sample					
Log(Sale Price)	12.1699	12.0844	12.3354	-	12.2292
Annualized Unlevered Returns	0.0840	0.0685	0.0646	-	0.0712
Annualized Levered (80%) Returns	0.3024	0.2237	0.2062	-	0.2389
Holding Length (Years)	5.3427	5.7665	5.9889	-	5.7456
Log(Purchase Price)	11.9294	11.8477	12.0807	-	11.9828
Sample Size	2,666,894	2,021,915	4,244,322	-	8,933,131

(a) Table I: Summary Statistics

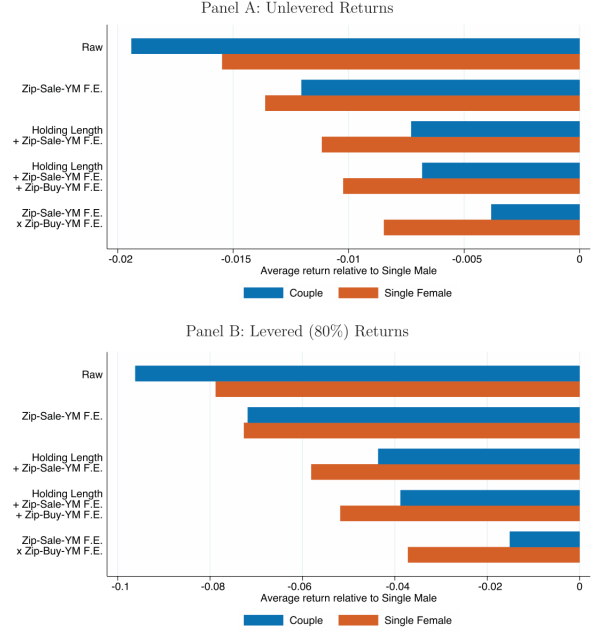


Figure 1. Annualized returns by gender group across different specifications. This figure plots the difference in annualized returns between single women and couples, relative to single men, conditional on the set of control variables listed on the y-axis. Panels A and B correspond to the coefficient estimates in Table II, Panels A and B, respectively. The levered returns are calculated following the formula in Section 1D, assuming an initial LTV of 80%. See Section 1B for more details on the definition of gender groups (single female, single male, and couple). (Color figure can be viewed at wileyonlinelibrary.com)

(b) Figure 1: Annualized returns by gender group

6.2 Table II and Figure 2: market timing and the return distribution

Read:

- Table II is the core return-regression table. In the raw specification, single women underperform single men by 1.5 percentage points in unlevered annualized returns.
- After zip-sale-year-month, zip-buy-year-month, and sale-year-month by buy-year-month fixed effects, the single-female gap remains -0.8 percentage points.
- Couples look worse than single women in raw returns, but most of their underperformance is explained by market timing.
- In levered returns, the raw female gap is -7.9 percentage points, and the fully controlled gap remains -3.7 percentage points.
- Figure 2 shows that men outperform women across much of the distribution, especially in the right tail.

Economic message: Market timing explains an important part of the gap, but not all of it. The remaining gap requires looking at transaction execution prices.

Table II
Housing Returns: Market Timing

This table reports the difference in annualized returns for single women and couples relative to single men (the omitted category). The levered returns are calculated following the formula in Section I.D, assuming an initial LTV of 80%. Columns (2) through (5) introduce additional control variables for the location and timing of housing transactions. Standard errors are clustered by zip code. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

Panel A: Unlevered Returns					
	Unlevered Ann Return				
	(1)	(2)	(3)	(4)	(5)
Single Female	-0.015*** (0.000)	-0.014*** (0.000)	-0.011*** (0.000)	-0.010*** (0.000)	-0.008*** (0.000)
Couple	-0.019*** (0.000)	-0.012*** (0.000)	-0.007*** (0.000)	-0.007*** (0.000)	-0.004*** (0.000)
Holding Length			-0.006*** (0.000)	-0.001 (0.001)	
Zip-SaleYM FE	No	Yes	Yes	Yes	Yes
Zip-BuyYM FE	No	No	No	Yes	Yes
SaleYM FE × BuyYM FE	No	No	No	No	Yes
R ²	0.004	0.352	0.378	0.534	0.592
Observations	8,933,131	8,933,131	8,933,131	8,933,131	8,933,131

Panel B: Levered (80% LTV) Returns					
	Levered (80%) Ann Return				
	(1)	(2)	(3)	(4)	(5)
Single Female	-0.079*** (0.001)	-0.073*** (0.001)	-0.058*** (0.001)	-0.052*** (0.001)	-0.037*** (0.001)
Couple	-0.096*** (0.002)	-0.072*** (0.001)	-0.044*** (0.001)	-0.039*** (0.001)	-0.015*** (0.001)
Holding Length			-0.037*** (0.000)	-0.026*** (0.006)	
Zip-SaleYM FE	No	Yes	Yes	Yes	Yes
Zip-BuyYM FE	No	No	No	Yes	Yes
SaleYM FE × BuyYM FE	No	No	No	No	Yes
R ²	0.004	0.292	0.326	0.481	0.628
Observations	8,933,131	8,933,131	8,933,131	8,933,131	8,933,131

(a) Table II: Housing Returns and Market Timing

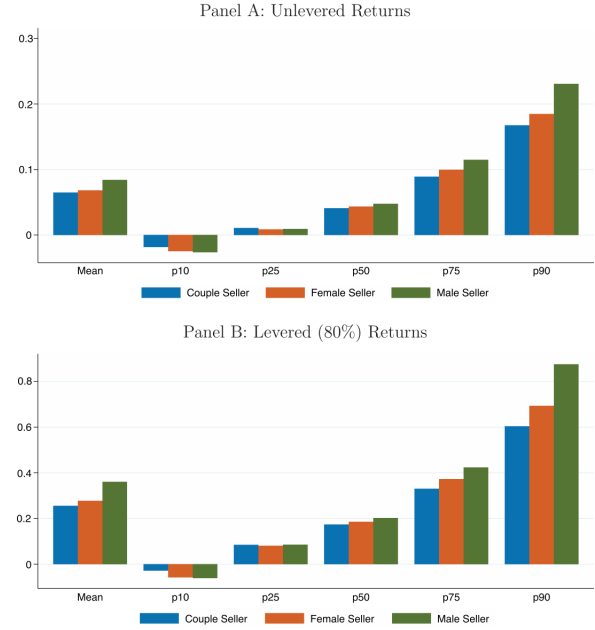


Figure 2. Distribution of annualized returns by gender group. This figure plots summary statistics for the annualized returns for housing transactions by three gender groups: couples, single women, and single men. The levered returns are calculated following the formula in Section I.D, assuming an initial LTV of 80%. (Color figure can be viewed at wileyonlinelibrary.com)

(b) Figure 2: Distribution of returns by gender group

6.3 Figure 3 and Table III: return-density patterns and demographic heterogeneity

Read:

- Figure 3 plots return densities by gender group. Men weakly outperform women across most of the distribution, while the left tail is not worse for men.
- The zoomed panel shows missing mass just to the left of zero, consistent with loss aversion and the disposition effect.
- Table III shows that the male-minus-female return gap is larger in zip codes with lower education, lower income, older residents, larger Black population share, and higher fraction single female.
- The gap remains positive even in high-income, high-education, and high-price zip codes.

Economic message: The gap is pervasive rather than confined to one demographic setting. It is larger in disadvantaged local areas but remains meaningful in high socioeconomic-status markets.

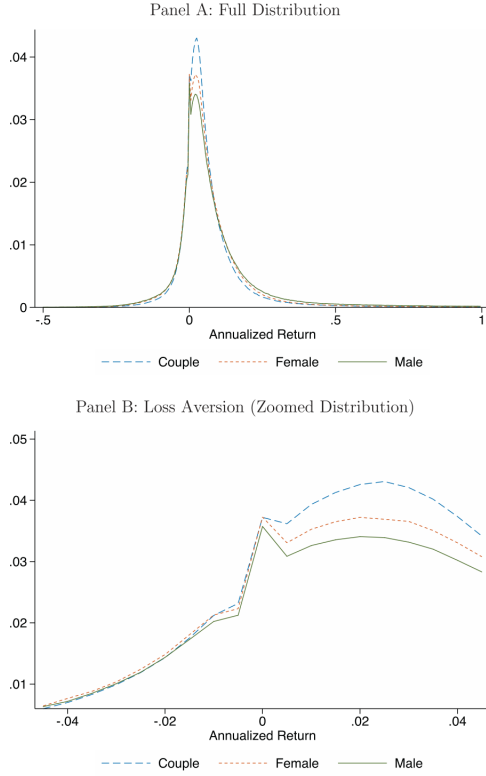


Figure 3. Density of unlevered annualized returns by gender group. This figure plots the density of the annualized unlevered returns for housing transactions by three gender groups: couples, single women, and single men. Returns are truncated at -50% and $+100\%$ in Panel A. Returns are truncated at -4% and $+5\%$ in Panel B. (Color figure can be viewed at wileyonlinelibrary.com)

Table III
Heterogeneity by Zip-Level Demographics: Quartile Averages

This table presents the average male minus female gender gap in unlevered annualized housing returns across quartiles of various zip-level demographic characteristics from the 2010 American Community Survey. The gender gap within each zip code is measured as the simple average across all transactions. Zip codes within each quartile are equally weighted to compute the quartile average.

Male - Female Unlevered Ann Return	Quartile 1 (1)	Quartile 2 (2)	Quartile 3 (3)	Quartile 4 (4)
Frac Black	0.0137	0.0140	0.0143	0.0194
Frac HS Education or Less	0.0089	0.0146	0.0196	0.0240
Frac 60+	0.0118	0.0154	0.0182	0.0184
Frac Single Female	0.0115	0.0135	0.0150	0.0222
Median Family Income	0.0234	0.0177	0.0145	0.0109
House Price	0.0191	0.0165	0.0139	0.0118

(a) Figure 3: Density of unlevered returns

(b) Table III: Zip-level demographic heterogeneity

6.4 Figures 4 and 5: geography and time-series persistence

Read:

- Figure 4 plots state-level male-minus-female unlevered return gaps. The gap is positive in almost all states with sufficient coverage.
- State-level variation may reflect data quality, measurement error, local institutions, housing-market structure, and genuine regional heterogeneity.
- Figure 5 shows average unlevered returns over time. Housing returns are procyclical and peak around the housing boom before the 2006 crash.
- The gender gap varies with the cycle but remains present across the sample and does not strongly disappear in recent years.

Economic message: The gender return gap is geographically broad and persistent over time, not a single-state or single-cycle artifact.

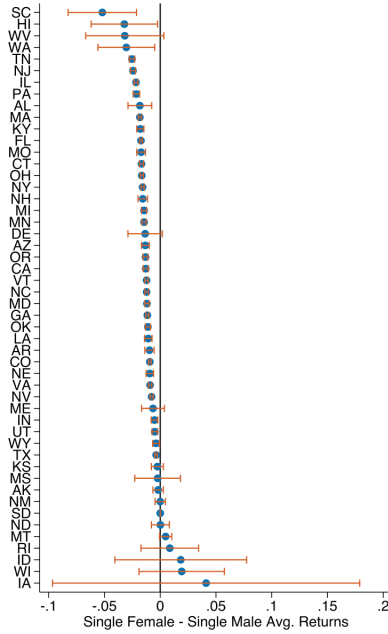


Figure 4. Gender gap in unlevered returns across states. This figure plots the average difference in unlevered annualized returns between single men and single women across states. The points represent the estimated difference in realized returns, while the bars represent the 95% confidence interval. Standard errors are clustered at the zip-code level. (Color figure can be viewed at wileyonlinelibrary.com)

(a) Figure 4: Gender gap across states

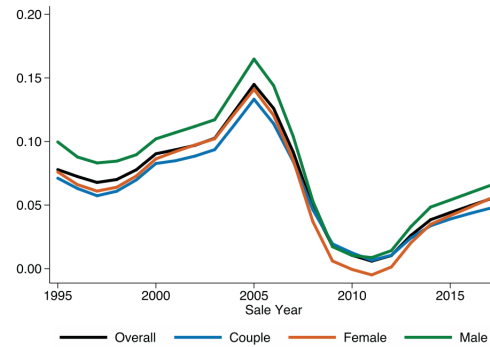


Figure 5. Average unlevered returns over time by gender group. This figure plots the average unlevered annualized return for couples, single women, and single men by sale year. As our sample begins in 1991, we begin this figure in 1995 to allow for sufficient data to avoid truncation. (Color figure can be viewed at wileyonlinelibrary.com)

(b) Figure 5: Average unlevered returns over time

6.5 Table IV and Figure 6: transaction prices and buyer-seller pairings

Read:

- Table IV compares transaction prices using property fixed effects and local time controls.
- Female buyers pay about 1.3% to 1.8% more than male buyers for the same property, depending on fixed effects.
- Female sellers sell the same property for about 2.6% to 2.9% less than male sellers.
- Couples pay more than single men when buying but sell for more than single men when selling.
- Figure 6 shows the gender-pairing result: male seller–female buyer pairs have the highest transaction prices; female seller–male buyer pairs have the lowest.

Economic message: The residual return gap is generated at both ends of the trade: women buy high and sell low, even when comparing the same property within the same local market-time cell.

Table IV
Transaction Price

This table examines variation in transaction prices across gender groups. Gender groups in columns (1) and (2) refer to buyers, and gender groups in columns (3) and (4) refer to sellers. We use repeat sales data that allow us to control for property fixed effects. We also control for zip-year-month fixed effects to account for time trends within a zip code. Each observation is a transaction. To better estimate property fixed effects, we do not restrict the sample to buyers or sellers with identified genders and matched names across sales and initial purchase. All observations corresponding to nonidentified single women, single men, and couples are included and coded as the "other" category. Single males are the omitted category. Standard errors are clustered by zip code. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

	Log(Purchase Price)		Log(Sale Price)	
	(1)	(2)	(3)	(4)
Single Female	0.013*** (0.001)	0.018*** (0.001)	-0.029*** (0.001)	-0.026*** (0.001)
Couple	0.029*** (0.002)	0.032*** (0.001)	0.014*** (0.001)	0.016*** (0.001)
Other	0.091*** (0.005)	0.022*** (0.003)	-0.051*** (0.002)	-0.051*** (0.001)
Property FE	Yes	Yes	Yes	Yes
Year-Month FE	Yes	No	Yes	No
Zip-Year-Month FE	No	Yes	No	Yes
R ²	0.794	0.886	0.793	0.886
Observations	52,883,866	52,883,866	52,883,866	52,883,866

(a) Table IV: Transaction Price

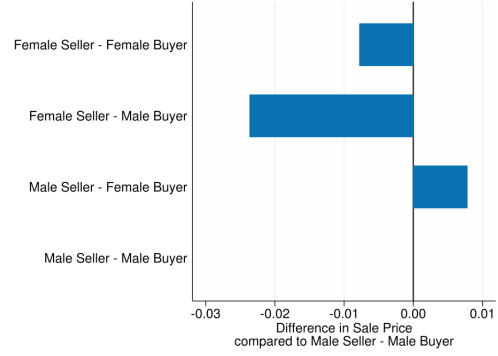


Figure 6. Transaction price by seller-buyer gender pairing. This figure shows the average difference in log transaction prices for each possible seller-buyer gender pair, relative to transactions involving single male sellers and single male buyers. These estimates come from a regression of the form in Table IV, column (4), but allow the buyer and seller gender group indicators to interact. We depict only the coefficients representing single male or single female buyers and sellers, with male seller-male buyer as the omitted base coefficient. (Color figure can be viewed at wileyonlinelibrary.com)

(b) Figure 6: Transaction price by seller-buyer pairing

6.6 Tables V and VI: list prices and negotiated discounts

Read:

- Table V shows that female buyers buy properties listed at higher prices, while female sellers list the same property at lower prices.
- Female sellers therefore do not simply sell low because they list high and then accept discounts; they start from a lower list price.
- Table VI shows that female buyers receive smaller purchase discounts than male buyers.
- Female sellers grant larger sale discounts than male sellers.
- Couples again sit between men and women when buying, but they negotiate especially favorable outcomes when selling.

Economic message: The gap operates through both first offers and bargaining. Women list at less favorable prices and also receive less favorable negotiated discounts.

Table V
List Price

This table examines variation in list prices across gender groups. Gender groups in columns (1) and (2) refer to buyers and gender groups in columns (3) and (4) refer to sellers. We use repeat sales data that allow us to control for property fixed effects. We also control for zip-year-month fixed effects to account for time trends within a zip code. Each observation is a listing matched to a sales transaction and the sample is restricted to properties for which we observe at least two transactions. To better estimate property fixed effects, we do not restrict the sample to buyers or sellers with identified genders and matched names across sales and initial purchase. All observations corresponding to nonidentified single women, single men, and couples are included and coded as the "other" category. Single males are the omitted category. Standard errors are clustered by zip code. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

	Log(Purchase List Price)		Log(Sale List Price)	
	(1)	(2)	(3)	(4)
Single Female	0.021*** (0.001)	0.022*** (0.001)	-0.021*** (0.001)	-0.018*** (0.000)
Couple	0.023*** (0.001)	0.015*** (0.000)	-0.008*** (0.001)	-0.004*** (0.001)
Other	-0.022*** (0.002)	-0.027*** (0.001)	-0.099*** (0.002)	-0.080*** (0.002)
Property FE	Yes	Yes	Yes	Yes
Year-Month FE	Yes	No	Yes	No
Zip-Year-Month FE	No	Yes	No	Yes
R ²	0.916	0.961	0.918	0.962
Observations	10,255,902	10,255,902	10,255,902	10,255,902

(a) Table V: List Price

Table VI
Discount Relative to the List Price

This table examines how negotiated transaction discounts vary by gender group. Gender groups in columns (1) and (2) refer to buyers and gender groups in columns (3) and (4) refer to sellers. We measure purchase and sale discounts as (list price - transaction price)/list price $\times$ 100, so a larger purchase discount contributes to a higher return on housing investment and a larger sale discount contributes to a lower return on housing investment. Standard errors are clustered by zip code. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

	Purchase Discount		Sale Discount	
	(1)	(2)	(3)	(4)
Single Female	-0.276*** (0.007)	-0.256*** (0.005)	0.167*** (0.007)	0.092*** (0.005)
Couple	-0.117*** (0.012)	-0.084*** (0.005)	-0.280*** (0.012)	-0.228*** (0.005)
Zip-Year-Month FE	No	Yes	No	Yes
R ²	0.001	0.222	0.002	0.257
Observations	13,690,139	13,690,139	9,658,833	9,658,833

(b) Table VI: Discount Relative to List Price

6.7 Figure 7 and Table VII: discount pairings and liquidity trade-offs

Read:

- Figure 7 shows that discounts are largest for female-seller/male-buyer pairs and smallest for male-seller/female-buyer pairs.
- This mirrors the transaction-price pairing evidence and is consistent with men receiving more surplus in mixed-gender bargaining pairs.
- Table VII asks whether women receive liquidity benefits in exchange for lower list prices and larger discounts.
- Female sellers have at most very small advantages in probability of sale and days on market, and these advantages shrink or flip after property fixed effects.
- The average economic magnitude is about one day faster sale time and less than one percentage point higher probability of sale.

Economic message: Lower sale prices for women are not offset by large liquidity benefits. The liquidity gains are too small to explain the return gap.

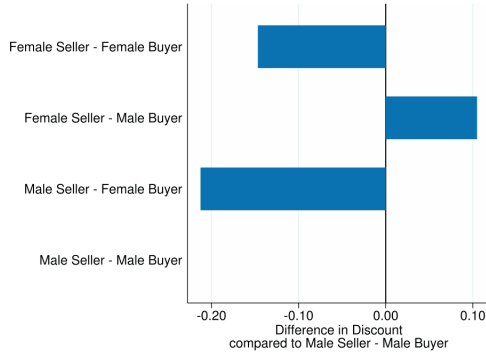


Figure 7. Discounts by seller-buyer gender pairing. This figure plots the average difference in discounts for each possible seller-buyer gender pair, relative to discounts for transactions involving single male sellers and single male buyers. We measure discounts as (list price – transaction price)/list price $\times 100$, so a larger discount contributes to a higher return on housing investment for buyers and a lower return for sellers. These estimates come from a regression of the form in Table VI, column (4), but allowing for the buyer and seller gender group indicators to interact. We plot only the coefficients representing single male or single female buyers and sellers, with male seller-male buyer as the omitted base category. (Color figure can be viewed at wileyonlinelibrary.com)

(a) Figure 7: Discounts by seller-buyer pairing

Table VII
Probability of Sale and Transaction Time

This table examines how owner gender relates to the probability of sale and days on market. Columns (1) to (4) examine the first listing after a purchase transaction in our sample, using the previous buyers' genders and family structure. Columns (1) and (2) examine whether the property listing is associated with a successful sale within our data sample. Columns (3) and (4) examine the listing's days on market, measured as the logarithm of the number of days between the initial list date and either the sale date or the listing withdrawal date. Columns (5) and (6) repeats the specification for columns (3) and (4), but looks at the days on market for all successful listings associated with a completed transaction within our sample, using the seller's gender and family structure. Standard errors are clustered by zip code. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

	Listing Pr(Sale)		Listing Lag(Days on Mkt)		Sale Lag(Days on Mkt)	
	(1)	(2)	(3)	(4)	(5)	(6)
Single Female	0.007*** (0.000)	-0.003*** (0.000)	-0.023*** (0.001)	0.003*** (0.001)	-0.027*** (0.002)	-0.009*** (0.002)
Couple	0.008*** (0.000)	0.005*** (0.000)	0.005*** (0.001)	-0.008*** (0.001)	-0.048*** (0.002)	-0.049*** (0.002)
Other	-0.019*** (0.000)	0.000 (0.000)	0.007*** (0.001)	0.008*** (0.001)	-0.002 (0.003)	-0.049*** (0.003)
Property FE	No	Yes	No	Yes	No	Yes
Zip-Year-Month FE	No	Yes	No	Yes	No	Yes

(b) Table VII: Probability of Sale and Transaction Time

6.8 Figure 8 and Table VIII: holding period and property controls

Read:

- Figure 8 shows that the annualized female-minus-male return gap is largest for short holding periods and decays toward zero with holding length.
- The gender gap in purchase and sale prices remains visible even for long holding lengths.
- This pattern fits the execution-price story: a fixed disadvantage at purchase and sale matters less per year when the holding period is long.
- Table VIII shows that adding property characteristics and property-type fixed effects does not eliminate the gender gap in annualized returns.
- Controls include age of unit, garage, pool, cooling, fireplace, basement, waterfront, bathrooms, square footage, bedrooms, upgraded status, new construction, and property type.

Economic message: The gap is not primarily about men buying observably different types of homes. Holding-length patterns strongly support fixed transaction-execution differences rather than annual risk premia or maintenance differences.

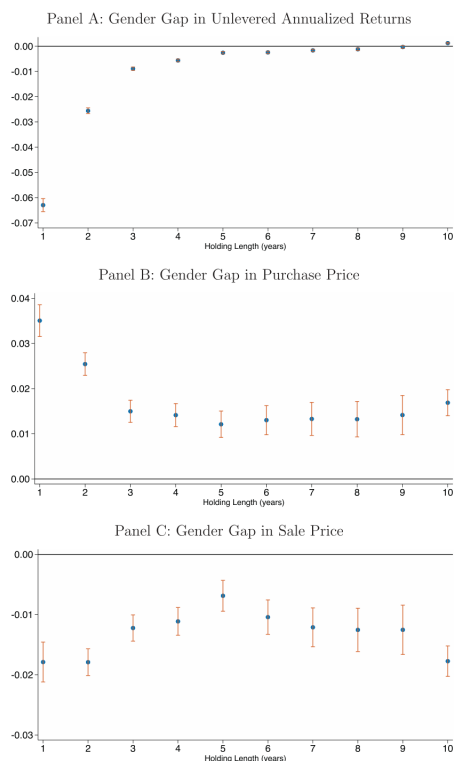


Figure 8. Differences between single men and single women by holding period. This figure plots the average difference in outcomes between single men and single women by holding length for the property. We exclude holding periods longer than 11 years. Panel A plots the female minus male gender gap in returns controlling for zip-sale-year-month fixed effects. Panels B and C plot the female minus male gender gap in log purchase and sale prices, respectively, controlling for property fixed effects and zip-year-month fixed effects. Vertical bars represent 95% confidence intervals. (Color figure can be viewed at wileyonlinelibrary.com)

(a) Figure 8: Differences by holding period

Table VIII
Controlling for Timing and Property Characteristics

This table examines how the gender gap in housing returns varies with additional control variables for market timing and property and listing agent characteristics. The sample is restricted to observations for which we have matched listings data on property characteristics. Standard errors are clustered by zip code. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

	Unlevered Ann Return		
	(1)	(2)	(3)
Single Female	-0.012*** (0.000)	-0.007*** (0.000)	-0.006*** (0.000)
Couple	-0.015*** (0.000)	-0.004*** (0.000)	-0.004*** (0.000)
Log(Age of Unit)			0.006*** (0.000)
Garage			-0.006*** (0.000)
Pool			-0.002*** (0.000)
Cooling			-0.002*** (0.000)
Fireplace			-0.004*** (0.000)
Basement			0.001** (0.000)
Waterfront			0.003*** (0.001)
Bathrooms			0.001 (0.001)
Log(Sq Ft)			0.003*** (0.001)
Bedrooms			0.001*** (0.000)
Upgraded			0.008*** (0.000)
New Construction			-0.004*** (0.000)
Property Type FE	No	No	Yes
Zip-SaleYM FE	No	Yes	Yes
Zip-BuyYM FE	No	Yes	Yes
SaleYM FE × BuyYM FE	No	Yes	Yes
R ²	0.003	0.646	0.649
Observations	3,008,113	3,008,113	3,008,113

(b) Table VIII: Timing and property characteristics

6.9 Table IX: AHS maintenance and demographic validation

Read:

- Table IX uses American Housing Survey data, where gender and household structure are self-reported.
- Column (1) shows no meaningful gender difference in reported annual maintenance scaled by purchase price.
- Columns (2) to (5) show that the gender gap in returns remains in AHS data, even with demographic controls.
- The gap persists after controlling for number of children, adults, education, race, family income, age of house, and MSA-by-year effects.
- Having children is associated with lower housing returns, but single women without children still earn lower returns than single men without children.

Economic message: The AHS evidence validates the main result in a data set with self-reported gender and richer demographics. Maintenance and observable demographics do not explain the whole gap.

Table IX
Maintenance and Housing Returns in the American Housing Survey

This table uses data from the American Housing Survey. In column (1), the dependent variable is reported annual maintenance scaled by home purchase price. In column (2), estimated unlevered return is the annualized unlevered return, calculated using the homeowner's self-reported estimate of current home value relative to purchase price. In columns (3) to (5), real unlevered return is the annualized unlevered return, calculated using the actual purchase price and sale price. Standard errors are double clustered by household and survey year. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

	Maint/price (1)	Est Unlevered Return (2)	Real Unlevered Return		
			(3)	(4)	(5)
Single Female	-0.000233 (0.000159)	-0.00595** (0.00241)	-0.0120** (0.00509)	-0.0102** (0.00511)	
Couple	0.000372*** (0.000133)	-0.00784*** (0.00213)	-0.00634 (0.00506)	-0.00577 (0.00479)	
Age of House	0.0000975*** (0.00000210)	0.000367*** (0.0000310)	0.000228*** (0.0000477)	0.000228*** (0.0000476)	0.000224*** (0.0000476)
Age of Householder	0.0000469*** (0.00000322)	-0.000617*** (0.0000529)	-0.0000800 (0.0000734)	-0.000140 (0.0000935)	-0.000146 (0.0000974)
Single Female with Child			-0.0193** (0.00776)		
Single Female w/o Child			-0.0103* (0.00576)		
Male with Child			-0.0124 (0.00858)		
Couple with Child			-0.0105 (0.00643)		
Couple w/o Child			-0.00665 (0.00567)		
Number of Children				-0.00293** (0.00123)	
Number of Adults				-0.00231	-0.00197

(a) Table IX, part 1: Maintenance and AHS returns

Table IX—Continued

	Maint/price (1)	Est Unlevered Return (2)	Real Unlevered Return		
			(3)	(4)	(5)
Some College				0.00145 (0.00086)	0.00156 (0.00086)
College Degree				0.00101 (0.00020)	0.000987 (0.00021)
Graduate Degree				-0.000606 (0.00365)	-0.000769 (0.00365)
Black				-0.00689 (0.00627)	-0.00737 (0.00631)
American Indian				-0.0114 (0.00941)	-0.0114 (0.00971)

(b) Table IX, part 2: continued controls

6.10 Tables X and XI: listing agents and market tightness

Read:

- Table X adds listing-agent fixed effects. The gender gaps in returns, discounts, list prices, and transaction prices remain similar.
- This implies that the gap is not primarily due to women hiring worse listing agents.
- The test cannot rule out gender-specific advice from the same agent or gender differences in whether clients follow agent advice.
- Table XI interacts gender indicators with market tightness, defined as sales relative to outstanding listings in a county-month.
- As market tightness rises, the female return gap, purchase discount gap, sale discount gap, and price gaps all shrink.

Economic message: Agent quality does not explain the gap. The market-tightness result is one of the strongest pieces of evidence for negotiation: when markets become auction-like and less bilateral, the gender gap narrows sharply.

Table X
Controlling for Listing Agents

This table reestimates the main regressions from Tables II, IV, V, and VI, controlling for listing agent fixed effects. The sample is restricted to observations with matched listing agent data. Agents are identified using their listing agent code or a listing matched to a transaction. Standard errors are clustered by zip code. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

	Unlevered Ann Return (1)	Purchase Discount (2)	Sale Discount (3)	Log(Purchase List Price) (4)	Log(Sale List Price) (5)	Log(Purchase Price) (6)	Log(Sale Price) (7)
Single Female	-0.012*** (0.000)	-0.233*** (0.005)	0.082*** (0.004)	0.013*** (0.000)	-0.013*** (0.000)	0.016*** (0.001)	-0.024*** (0.001)
Couple	-0.019*** (0.000)	-0.042*** (0.004)	-0.175*** (0.004)	0.019*** (0.000)	0.004*** (0.000)	0.023*** (0.001)	0.014*** (0.001)
Other		0.431*** (0.006)	0.246*** (0.006)	-0.017*** (0.000)	-0.030*** (0.001)	0.017*** (0.003)	-0.042*** (0.001)
Property FE	No	No	No	Yes	Yes	Yes	Yes
Zip-Year-Month FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Listing Agent FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R ²	0.454	0.298	0.285	0.975	0.975	0.887	0.888
Observations	5,377,609	19,941,435	19,941,435	10,210,719	10,210,719	28,673,949	28,673,949

(a) Table X: Controlling for Listing Agents

Table XI
Variation by Market Tightness

This table reestimates the main regressions from Tables II, IV, and VI, after interacting the gender group indicators with a measure of market tightness. Market tightness is defined as the number of sales in a given county-month, scaled by the outstanding number of listings currently for sale in that county-month. Standard errors are clustered by zip code. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

	Unlevered Ann Return (1)	Purchase Discount (2)	Sale Discount (3)	Log(Purchase Price) (4)	Log(Sale Price) (5)
Single Female	-0.017*** (0.000)	-0.284*** (0.019)	0.039*** (0.014)	0.024*** (0.001)	-0.029*** (0.001)
Couple	-0.012*** (0.000)	-0.249*** (0.018)	-0.012 (0.014)	0.013*** (0.001)	0.017*** (0.001)
Other		0.014 (0.016)	0.349*** (0.016)	0.032*** (0.002)	-0.060*** (0.002)
Single Female × Tightness	0.020*** (0.002)	0.284*** (0.095)	-0.487*** (0.060)	-0.039*** (0.004)	0.019*** (0.004)
Couple × Tightness	-0.001 (0.002)	0.413*** (0.091)	-0.026 (0.065)	-0.004 (0.006)	-0.024*** (0.005)
Other × Tightness		0.068 (0.077)	0.135* (0.074)	-0.006 (0.008)	0.059*** (0.009)
Property FE	No	No	No	Yes	Yes
Zip-Year-Month FE	Yes	Yes	Yes	Yes	Yes

(b) Table XI: Variation by Market Tightness

7 Short summary

- The paper studies gender differences in realized housing returns using U.S. deeds and listings data from 1991 to 2017.
- Single women earn about 1.5 percentage points lower unlevered annualized returns than single men.

- The gap becomes much larger under mortgage leverage: assuming an 80% LTV mortgage, the raw levered gap is about 7.9 percentage points per year.
- Around 45% of the raw unlevered gap is explained by market timing and location.
- The remaining gap is linked to execution prices: women buy the same property for more and sell it for less.
- Linked MLS data show that women list for less and receive worse discounts relative to the list price.
- Female seller–male buyer transactions generate especially low prices; male seller–female buyer transactions generate especially high prices.
- Female sellers do not receive large liquidity benefits in exchange for lower prices.
- The annualized return gap decays toward zero with holding length, consistent with fixed execution-price disadvantages at purchase and sale.
- Property characteristics, maintenance, upgrades, demographics, and listing-agent quality do not explain the gap.
- The gender gap shrinks in tight markets, where bilateral negotiation is replaced by quasi-auction bidding.
- The paper estimates that housing-return differences explain roughly 30% of the gender gap in wealth accumulation at retirement.

8 Conclusion

8.1 Core conclusion

Goldsmith-Pinkham and Shue show that housing markets generate a large and economically important gender gap in investment returns. Single women earn lower housing returns than single men, and the gap is meaningful for wealth accumulation because housing is the dominant asset for many households and is often purchased with leverage.

8.2 Economic intuition

Housing differs from stocks and bonds because it is heterogeneous, illiquid, and negotiated. Two investors buying shares of the same index fund at the same time receive the same price, but two investors buying or selling a house can receive different prices for the same property. This creates room for gender differences in timing, first offers, search, negotiation, counterparties, and transaction execution.

8.3 Incremental contribution revisited

The paper’s central contribution is not only documenting a gender gap in housing prices. It measures realized returns, decomposes the gap into market timing and execution, and then uses linked listing data to show that the execution component is visible in list prices and negotiated discounts. The paper therefore connects household finance, real estate, negotiation, and gender inequality.

8.4 Implications

- **Household finance.** Housing-return differences matter for wealth inequality because housing is the main asset of most households.
- **Real estate market design.** Posted-price or quasi-auction mechanisms may reduce gender gaps by reducing the role of bilateral negotiation.
- **Financial advice.** Advice about list prices, reservation prices, and negotiation strategy may have large wealth effects.
- **Gender inequality.** The paper shows that gender gaps in wealth accumulation can arise outside the labor market through asset-market transactions.

8.5 Limitations

- The paper measures financial returns, not welfare or consumption value from owning a home.
- Gender and family structure are inferred from names in the main data, though AHS validation mitigates this concern.
- The paper cannot fully distinguish negotiation ability, negotiation preferences, discrimination, advice, time constraints, and search intensity.
- Agent fixed effects rule out fixed differences in average agent quality, but not gender-specific advice or differences in how clients respond to advice.

8.6 Takeaway

The paper's main takeaway is that gender gaps in asset returns are not limited to financial securities. In housing - a large, leveraged, negotiated asset - women receive lower realized returns because of both market timing and less favorable execution prices at purchase and sale. These differences are large enough to matter for long-run wealth accumulation.